

Information, KL-divergence, Minimum Description Length

## Communicating Via a Noisy Channel

- Information theory considers communicating down a (noisy) channel

$$X \sim \mathbb{P}(X) \xrightarrow{\text{noisy channel}} Y \sim \mathbb{P}(Y | X)$$

- We send a message  $X$  (with probability  $\mathbb{P}(X)$ ) and receive a message  $Y$  with probability  $\mathbb{P}(Y | X)$
- The uncertainty of the message sent, given we received a message  $y$  is

$$H_{X|Y=y} = - \sum_{x \in \mathcal{X}} \mathbb{P}(X = x | Y = y) \log(\mathbb{P}(X = x | Y = y))$$

- The expected uncertainty in the message sent is

$$H_{X|Y} = \sum_{y \in \mathcal{Y}} \mathbb{P}(Y = y) H_{X|Y=y} = - \sum_{x,y} \mathbb{P}(X = x, Y = y) \log(\mathbb{P}(X = x | Y = y))$$

## Mutual Information

- The amount of uncertainty about the message being sent,  $X$ , before receiving the message is  $H_X = -\mathbb{E}_X[\log \mathbb{P}(X)]$
- Shannon define the *mutual information* to be the expected loss in uncertainty when we receive a message

$$I_{X;Y} = H_X - H_{X|Y}$$

- Since  $H_X - H_{X|Y} = H_Y - H_{Y|X}$  it follows

$$I_{X;Y} = I_{Y;X}$$

## Independent Noise

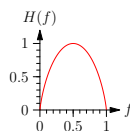
- The simplest model of a noisy channel is a binary channel where each symbol is corrupted independently with a probability  $f$

$$\mathbb{P}(X = 1 | Y = 0) = \mathbb{P}(X = 0 | Y = 1) = f$$

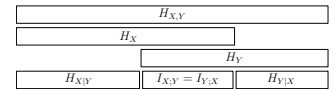
- An elementary calculations shows that

$$H_{X_i|Y_i} = -(1-f) \log(1-f) - f \log(f) = H(f)$$

- For a message of length  $n$ ,  $H_{X|Y} = nH(f)$



- Information Theory
- KL-Divergence
- Minimum Description Length
- Variational Auto-Encoders



## Joint Entropy

- We can define the **joint entropy**

$$H_{X,Y} = - \sum_{x,y} P_{X,Y}(x,y) \log(P_{X,Y}(x,y))$$

- If the message we receive is independent of the message that is sent then  $H_{X,Y} = H_X + H_Y$  (we saw this in the last lecture)

- $H_{X,Y} \neq H_X + H_Y$  if  $X$  and  $Y$  are correlated

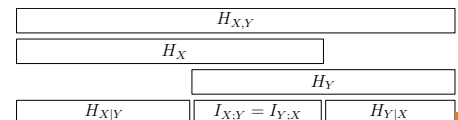
- Since  $\mathbb{P}(X,Y) = \mathbb{P}(Y|X)\mathbb{P}(X) = \mathbb{P}(X|Y)\mathbb{P}(Y)$  it follows

$$H_{X,Y} = H_X + H_{Y|X} = H_Y + H_{X|Y}$$

- Or  $H_X - H_{X|Y} = H_Y - H_{Y|X}$

## Channel Capacity

- We can summarise these relationships diagrammatically



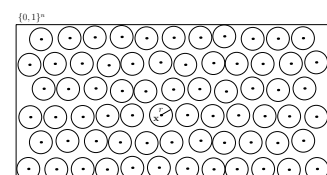
- Shannon defined the *capacity* of a noisy channel as

$$C = \max_{\mathbb{P}(X)} I_{X;Y}$$

- That is, you choose the probability distribution of the message to maximise the information gain

## Error Correcting Codes

- To reduce the chance of misinterpreting a message we need to build an error correcting code
- We can do this dividing the space of binary messages into a set of Hamming balls



- A Hamming ball  $B(x,r)$  is the set of strings that differ from  $n$ -dimensional binary string,  $x$ , by at most  $r$  digits

- The expected number of errors in a string of length  $n$  given an error rate of  $f$  is  $nf$
- For sufficiently large  $n$  we would expect all errors are smaller than  $(f + \epsilon)n$  (for  $\epsilon > 0$ )
- If we make the radius of the Hamming ball  $r = (f + \epsilon)n$  ( $\epsilon > 0$ ) then we would expect no error for sufficiently large  $n$
- An upper bound on the number of code words we can send in a string of length  $n$  is

$$\frac{2^n}{|B(x_i, r)|} = \frac{2^n}{\binom{n}{r}} = 2^n e^{-\log(n!) + \log(r!) + \log((n-r)!)} = c\sqrt{n}2^{I_{X;Y}}$$

## Using Mutual Information

- Mutual information is used quite often in machine learning
  - Wikipedia mentions 14 applications
- Suppose we want to align two sets of images through some non-linear transformations
- One way of doing this is to choose the non-linear transformations that maximise the mutual information (or normalised mutual information) between the two sets of images

## KL-Divergence

- We have met the Kullback-Leibler divergence
 
$$\text{KL}(p||q) = \mathbb{E}_{X \sim p(X)} \left[ \log \left( \frac{p(X)}{q(X)} \right) \right]$$

$$= \mathbb{E}_{X \sim p(X)} [-\log(q(X))] - H_X$$
- Recall  $-\log(q(X=x))$  is the length of code need to send a message  $x$  with a probability  $q(X=x)$
- Thus  $-\mathbb{E}_{X \sim p(X)} [\log(q(X))]$  is the expected length of message needed to code  $X \sim p(X)$  using the optimal code for the distribution  $q(X)$  rather than  $p(X)$
- $\text{KL}(p||q)$  is also known as the **relative entropy** and measures the expected extra length in coding  $X \sim p(X)$  if we use the wrong distribution  $q(X)$

## Evidence Lower Bound (ELBO)

- We can re-write the variational free energy as

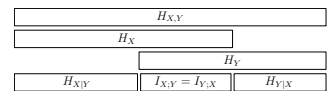
$$\begin{aligned} \Phi(\phi) &= - \int g(\theta | \phi) \log \left( \frac{g(\theta | \phi)}{(f(\theta, \mathcal{D})/f(\mathcal{D})) f(\mathcal{D})} \right) d\theta \\ &= - \int g(\theta | \phi) \left( \log \left( \frac{g(\theta | \phi)}{f(\theta | \mathcal{D})} \right) - \log(f(\mathcal{D})) \right) d\theta \\ &= -\text{KL}(g(\theta | \phi) || f(\theta | \mathcal{D})) + \log(f(\mathcal{D})) \end{aligned}$$

- If we maximise  $\Phi(\phi)$ , we end up minimising the KL divergence between  $g$  and  $f$  so that  $g \approx f$  and  $\Phi(\phi) \approx \log(f(\mathcal{D}))$
- That is, we choose the parameters of our simple factorised distribution so that it is close to the true posterior

- Shannon also showed that choosing  $2^{I_{X;Y}}$  random strings of length  $n$  the Hamming distance between balls would be at least  $f$  with high probability
- This means that we can send information at rate of  $I_{X;Y}$
- The maximum rate is given by the channel capacity  $\max I_{X;Y}$
- If  $f = 0.1$  then  $C = I_{X;Y} = 0.469$  bits so we need codes of just over twice as long to communicate accurately over a noisy channel with a 10% corruption rate
- Unfortunately, we can't efficiently decode random code positions, so although we know Shannon's bound is achievable we don't have practical codes that do this

## Outline

- Information Theory
- KL-Divergence
- Minimum Description Length
- Variational Auto-Encoders



## Variational Approximation

- Recall we use MCMC in Bayesian inference because the posterior distribution is too complicated to write down in closed form
- In the variational approximation we approximate the posterior distribution by a simpler (typically factored distribution), e.g.

$$f(\theta | \mathcal{D}) \approx g(\theta | \phi) = \prod_i g(\theta_i | \phi_i)$$

- The standard method for solving this is to maximise the **variational free energy**

$$\Phi(\phi) = - \int g(\theta | \phi) \log \left( \frac{g(\theta | \phi)}{f(\theta, \mathcal{D})} \right) d\theta$$

## Put Another Way

- We can rewrite the variational free energy as  $\Phi(\phi) = - \int g(\theta | \phi) \log \left( \frac{g(\theta | \phi)}{f(\theta, \mathcal{D})} \right) d\theta = L_g(\phi) + H_g(\phi)$  where

$$L_g(\phi) = \int g(\theta | \phi) (\log(f(\mathcal{D} | \theta) f(\theta))) d\phi$$

acts like an expected posterior that is maximised when the data is well modelled (we put  $g(\theta | \phi)$  where  $f(\theta, \mathcal{D})$  is large, c.f. MAP)

- The second term is an entropy

$$H_g(\phi) = - \int g(\theta | \phi) \log(g(\theta | \phi)) d\phi$$

That is, we maximise the uncertainty of the distribution  $g(\theta | \phi)$

- Variational methods can be much faster than MCMC (although they tend to involve some iterations to minimise the variation free energy)■
- They can produce very good approximations, although this is not guaranteed (depends on the problem)■
- They can be extended (e.g. by minimising  $\text{KL}(g\|f)$  rather than  $\text{KL}(f\|g)$ —this is known as *belief propagation*)■
- MCMC is less elegant, but is a controlled approximation (we get better results by increasing the number of iterations)■
- MCMC is slower, but on modern computers this isn't usually a problem■

## Compression and Model Selection

- Outside of the Bayesian framework it is difficult to do model selection—most of ML isn't Bayesian■
- When is it better to accept a more complex model for a better fit and when are we just over-fitting?■
- Usually we answer this using a validation set, but this is not always possible■
- One principled approach is to use the model that allows us to maximally compress the data■
- If we are compressing the data then we are capturing features of the data■

## Description Length

- The **description length** for  $\{y_i \mid i = 1, 2, \dots, m\}$  is then the cost of transmitting  $\theta$  plus the cost of transmitting the errors

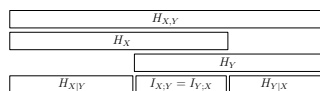
$$L = \sum_{k=1}^n \ell(\theta_k) - \sum_{i=1}^m \left( \log(p_\delta(y_i - \hat{f}(x_i|\theta))) \right) + \log(\Delta)$$

where  $\ell(\theta_k)$  is the number of bits need to communicate  $\theta_k$  (we get to choose the accuracy it is worth encoding the parameters)■

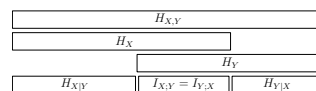
- To select between models we choose the model with the **minimum description length**■
- Note that the accuracy  $\Delta$  will lead to the same cost,  $-m \log(\Delta)$ , for all models so doesn't affect which model is selected■

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2. KL-Divergence
3. Minimum Description Length
4. **Variational Auto-Encoders**



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## Alice and Bob

- Suppose Alice has data  $\mathcal{D} = \{(x_i, y_i) \mid i = 1, 2, \dots, m\}$  while Bob has only the feature vectors  $\{x_i \mid i = 1, 2, \dots, m\}$ ■
- Alice wants to communicate  $y_i$  to Bob as efficiently as possible■
- We suppose Alice & Bob have available a model  $\hat{f}(x|\theta)$ ■
- Rather than sending the complete list  $\{y_i \mid i = 1, 2, \dots, m\}$  Alice can send Bob the parameter  $\theta$  and the errors

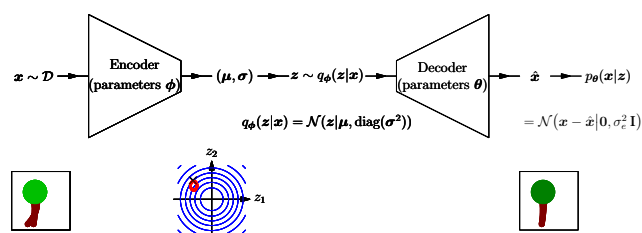
$$\delta_i = y_i - \hat{f}(x_i|\theta)$$

- Assuming the  $\delta_i$ 's have a distribution  $p_\delta$  then the cost of communicating an error to accuracy  $\Delta$  is  $-\log(p_\delta(\delta_i) \times \Delta)$ ■

## Minimum Description Length (MDL) Method

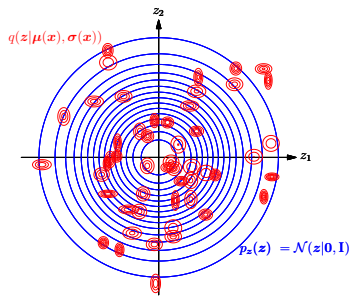
- The minimum description length method can be a powerful way of choosing between models■
- Often it is the only principled method available■
- It allows you to trade model accuracy against model complexity■
- It can be fiddly as we need to determine the accuracy to which we should store the parameters of our model■
- This isn't something we usually think about, but often we can get very good models even when we truncate the parameters to low precision■

## Variational Auto-Encoders VAE



$$\mathcal{L} = \mathbb{E}_{x \sim \mathcal{D}} [\text{KL}(q_\theta(z|x) \parallel \mathcal{N}(0, \mathbf{I})) - \log(p_\theta(x|z(x)))]$$

## Latent Space



$$\text{KL}(q_\theta(z|x) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I}))$$

## Understanding the Loss Function

- The original paper derived the loss function as a variational approximation to maximising some posterior
- This is difficult to understand (at least, for me)
- It has a very natural explanation in terms of minimum description length
- Alice wants to communicate the images to Bob
- Alice uses the encoder to derive a (latent) code  $q(z|x)$  which she communicates to Bob
- She also communicates the errors  $\delta = x - \bar{x}$
- Bob uses the decoder to decode  $q(z|x)$  and  $\delta$  to repair the images

## Description Length

- The loss

$$\mathcal{L} = \mathbb{E}_{x \sim \mathcal{D}} [\text{KL}(q_\theta(z|x) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})) - \log(p_\theta(x|z(x)))]$$

can be interpreted as

- ★ The cost of communicating the code  $\text{KL}(q_\theta(z|x) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I}))$
- ★ Plus the cost to send the repair  $\log(p_\theta(x|z(x)))$
- We minimise the loss function equivalent to MDL
- What is really clever is that we can choose the accuracy of the code we send  $q_\theta(z|x)$  to minimise the over-all cost

## Conclusions

- Information theory has regularly been used in machine learning
- It requires some understanding and care to do it properly
- The KL-divergence (or relative entropy) is often used to make two probability distribution more alike
- The minimum description length is a powerful principle for model selection
- Variational Auto-Encoders have a very natural interpretation in terms of minimising a description length